

Day - 65

Transformers

[Transformer] →
NN arch

seq 2 seq task

machine translation

question ans

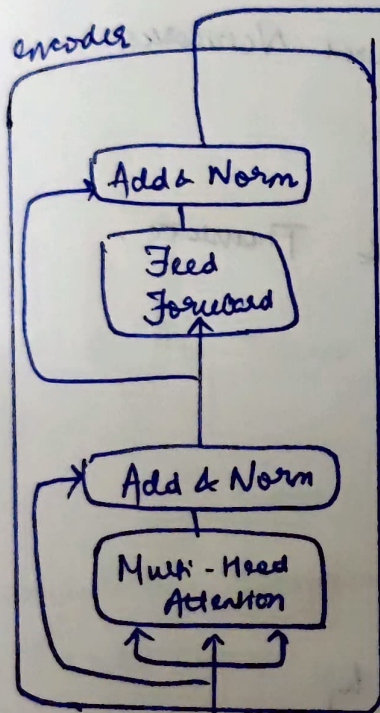
text summarization

self attention (parallel processing)

Softmax

Output-
Probabilities

Linear

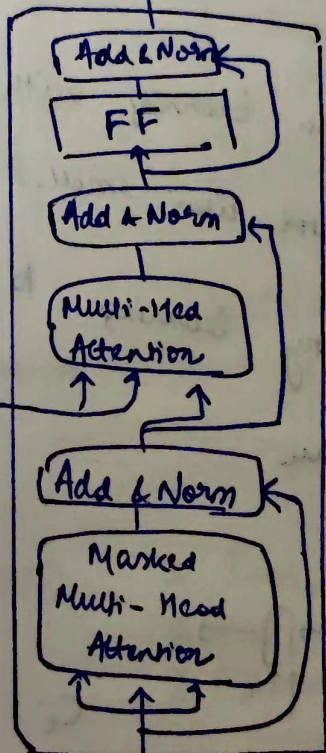


Positional
Encoding

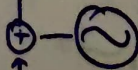


Input
Embedding

Inputs



N_2



Output
Embedding

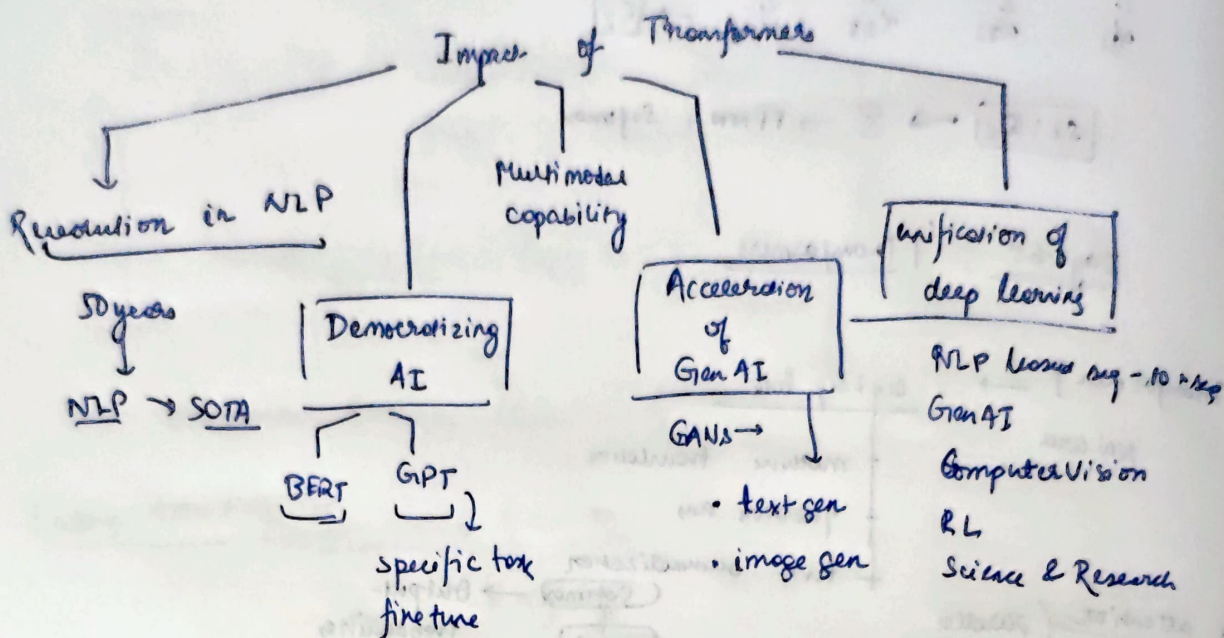
Outputs
(shifted right)

Positional
Encoding

Attention is All you Need, 2017

Asish Vaswani

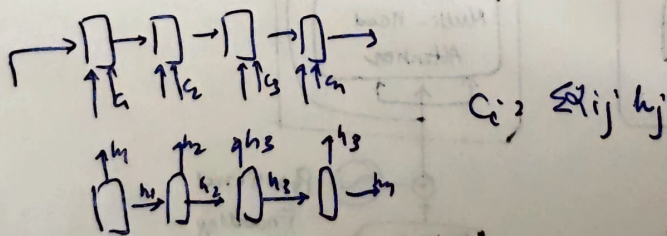
Niki Parmar



The origin story

①. Sequence to Sequence Learning with Neural Networks.
→ Context-recall was small.

②. NMT by jointly learning to Align & Translate.
Dmitry Bahdanau



Problem - Sequential training, (Slow) → not huge datasets.

So, we can't use transfer learning.

No pre-training → no fine-tuning

③ Attention is All you Need

→ LSTM/RNN not used, Self Attention

→ Stable

→ hyperparameters → stable/robust

2000 - 2014 → RNNs/LSTMs

2014 → Attention

2017 → Transformers

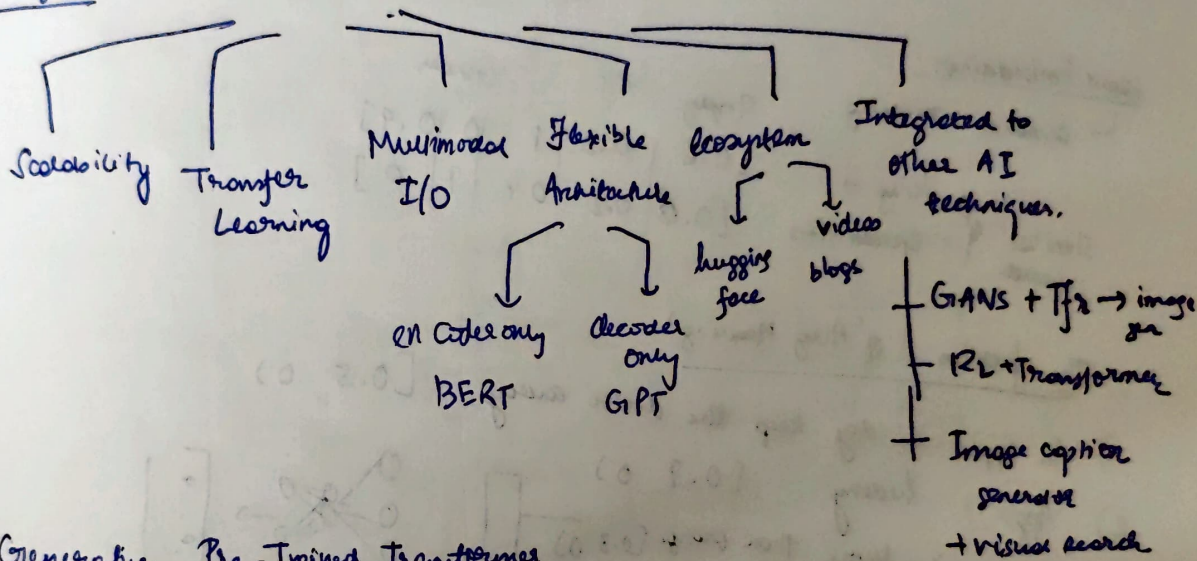
2019 → BERT / GPT (TL)

2018 - 2020 → Vision Transformer / AlphaFold-2

2021 → GenAI, GPT-3, DALL-E

2022 → CGPT / Stable Diffusion.

Advantages



Generative Pre-Trained Transformers.

Disadvantages

1. High computational resources. (GPUs)
2. Data (LLMs)
3. Risk of overfitting
4. Energy consumption.
5. Interpretability.
6. Bias.
7. Ethical concerns.

Future?

1. Improvements in efficiency.
2. Multimodal capabilities. — text; image/sound; biometric features.
3. Responsible development
4. Domain Specific
5. Multi Linguines.
6. Interpretability. → how did the results come?

Lec-72 | What is Self Attention

The What?

NLP Application - imp. requirement

NLP $\rightarrow$ words $\rightarrow$ numbers vectorization

① ONE - First technique

② Bag of words.

man cat man

$P \rightarrow [2 \ 0 \ 1]$

Word Embeddings

$\hookrightarrow$ Semantic meaning.

Similar words

king $\rightarrow$
queen $\rightarrow$

Roasting

$[0.6 \ 0.2 \ 1 \ 0 \ 0.9]$

$[0.6 \ 0.2 \ 0.4 \ 1 \ 0]$

Gender

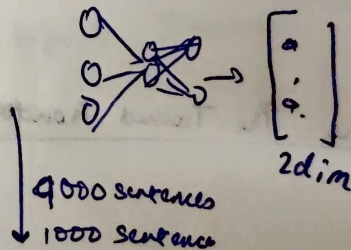
The problem of 'Avg Meaning'

1) An apple a day keep the doctor away $[0.5 \ 0]$

2) Apple is healthy $[0.7 \ 0]$

3) Apple is better than orange $[0.8 \ 0]$

4) Apple makes great phones. $[0.3 \ 0.2]$



$\Rightarrow [0.9 \ 0.3]$

Word Embedding depends on dataset.

$\downarrow$
Create once $\rightarrow$ use everywhere (static)

$\downarrow$ we need
(Contextual embedding)

[Apple launched a new Phone
while I was using an old one]

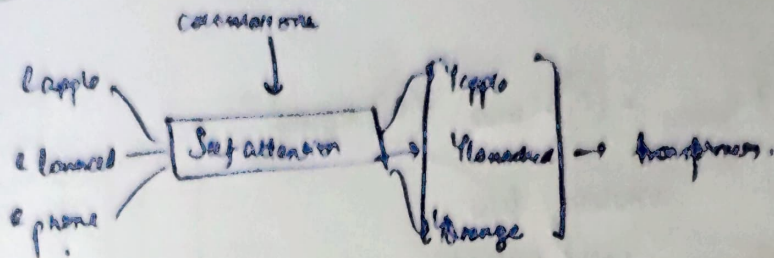
$[0.9 \ 0.3]$

$\downarrow \downarrow$
 $[0.3 \ 0.8]$

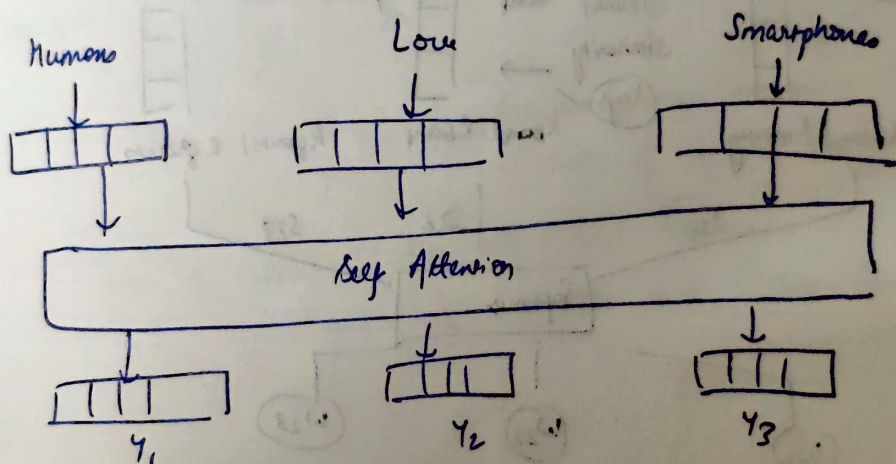
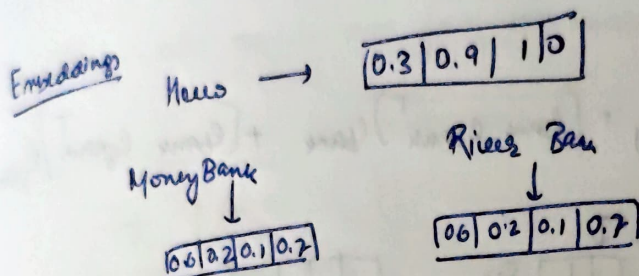
static
embedding

very

smart
contextual embedding.



Lee-73 | Self Attention in Transformers



First-Principle Approach

money bank grows

$$\begin{aligned} \text{bank} &\rightarrow \text{bank} \\ \text{bank} &\rightarrow 0.3 \text{ money} + 0.7 \text{ bank} \\ &\quad + 0.1 \text{ grows} \end{aligned}$$

rivers bank flows

$$\begin{aligned} \text{bank} &\rightarrow 0.5 \text{ rivers} + 0.4 \text{ bank} \\ &\quad + 0.1 \text{ flows} \end{aligned}$$

S_1

$$\begin{aligned} m &= 0.7m + 0.2b + 0.1g \\ b &= 0.25m + 0.7b + 0.05g \\ g &= 0.1m + 0.2b + 0.7g \end{aligned}$$

S_2

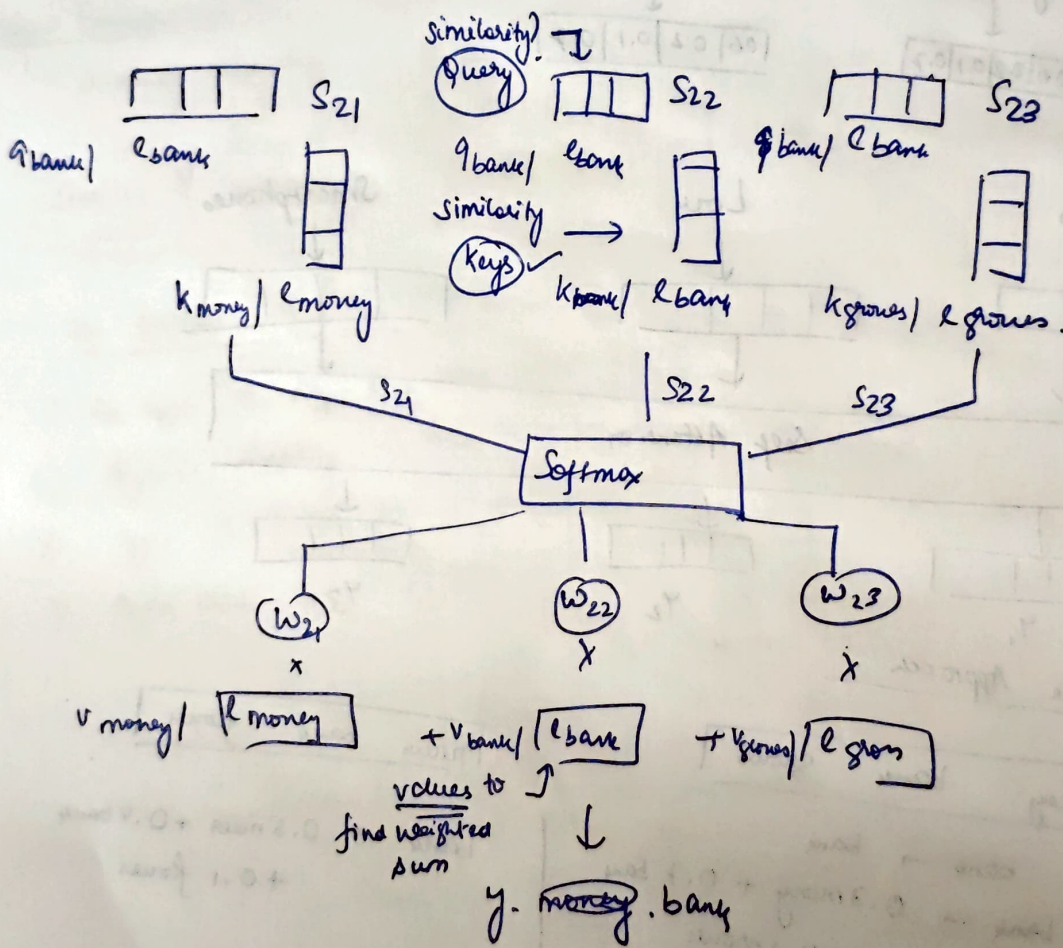
$$\begin{aligned} r &= 0.8r + 0.15b + 0.05f \\ b &= 0.2r + 0.78b + 0.02f \\ f &= 0.4r + 0.01b + 0.59f \end{aligned}$$

$$e_{\text{money}}^{\text{new}} = 0.7 e_{\text{money}} + 0.2 e_{\text{bank}} + 0.1 e_{\text{grows}}$$

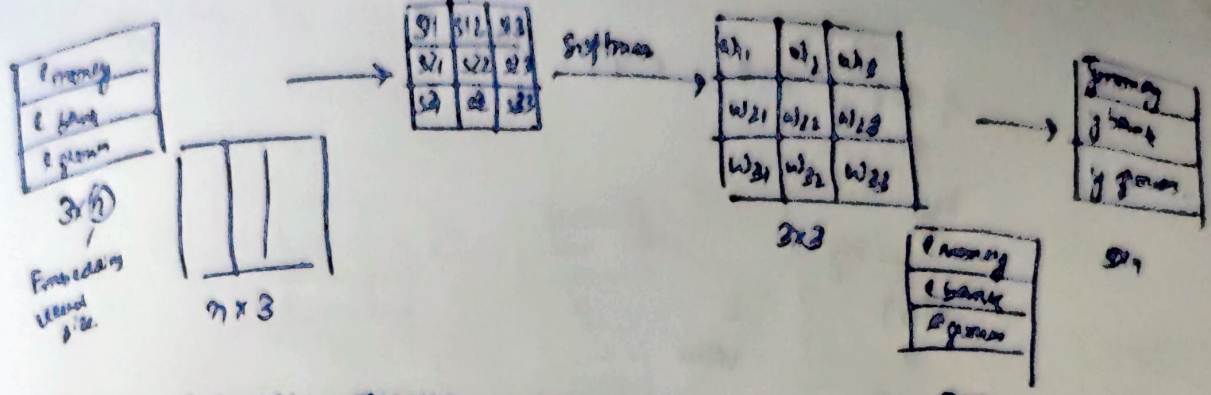
similarity le/w
word embedding

$$\begin{matrix} 0.7 & e_{\text{money}} & e_{\text{money}} \\ 0.2 & e_{\text{money}} & e_{\text{money}} \end{matrix} \quad \left(\begin{matrix} 1 & 2 & 3 \\ \text{money} & \text{bank} & \text{grows} \end{matrix} \right)$$

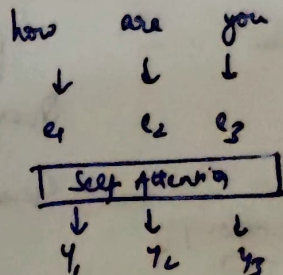
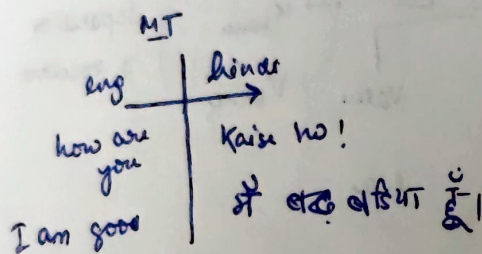
$$e_{\text{bank}}^{(\text{new})} = [e_{\text{bank}} \cdot e_{\text{money}}^T] e_{\text{money}} + [e_{\text{bank}} \cdot e_{\text{bank}}^T] e_{\text{bank}} + [e_{\text{bank}} \cdot e_{\text{grows}}^T] e_{\text{grows}}$$



parallelly calculate embedding.



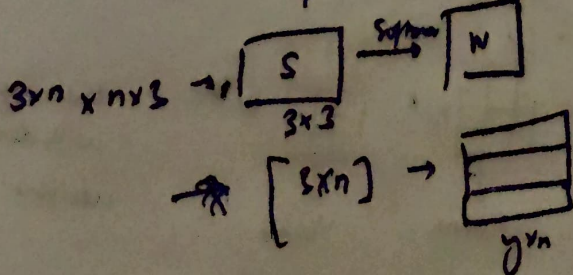
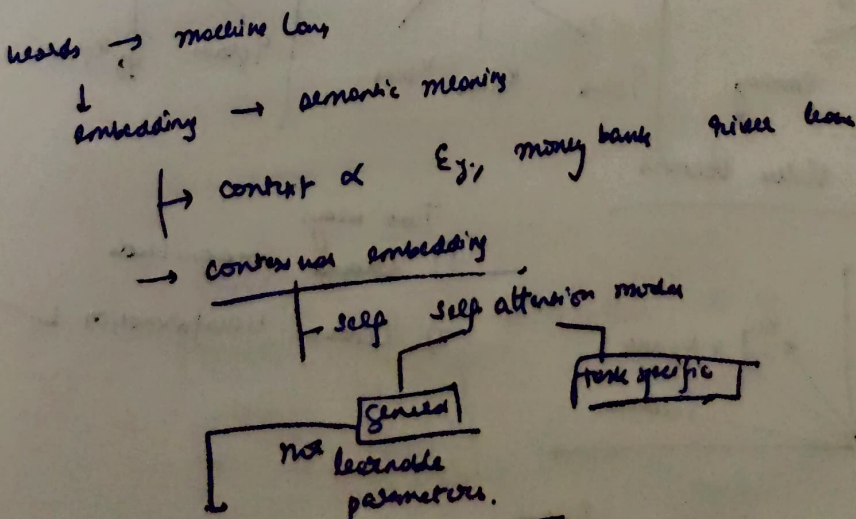
→ Not a sequential operation → parallel operation
 → There are no learning parameters involved.

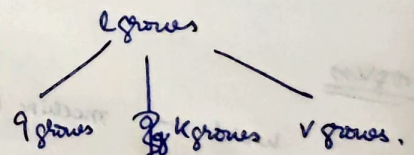
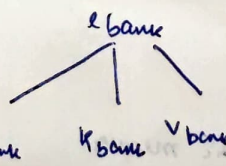
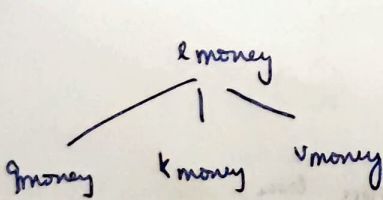
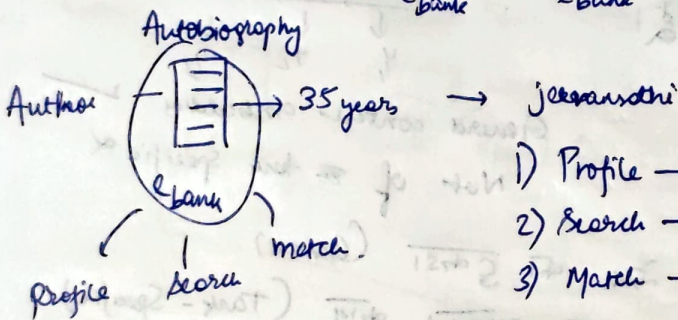
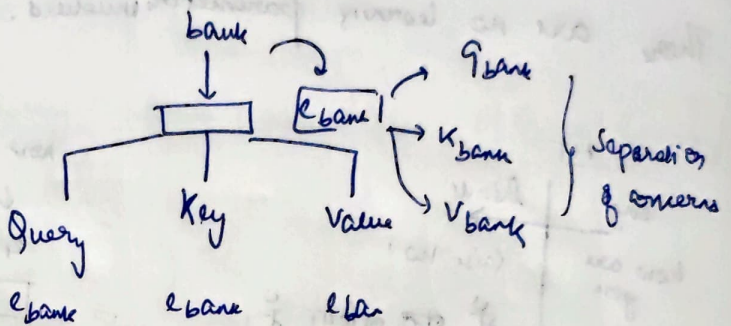
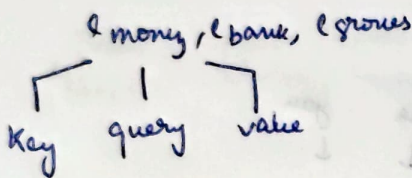
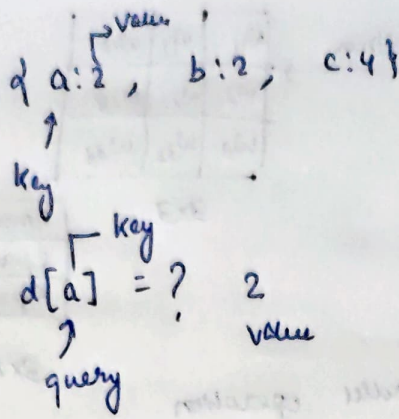


General context embeddings
 Not of task specific

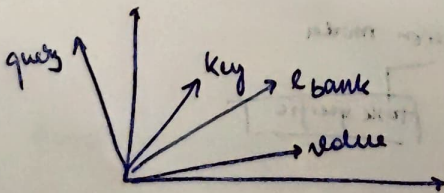
piece of cake — $\frac{1}{\sqrt{d}}$ (General)
 → $\frac{1}{\sqrt{d}}$ आसान काम (Task-Specific)

Progress

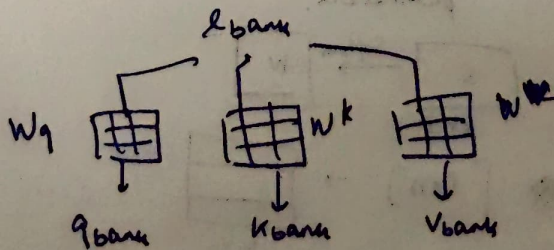




Query, Key & Value Vectors



- Two ways
- 1) Change magnitude
 - 2) Linear transformation



training process we'll decide the weight matrices.

w_q, w_k, w_v is same for all the embedding vectors.

